

Project Title:

Healthcare Medical Data — Power BI Report:

Project Objective:

The main objective of the **Medical Healthcare Data Analysis project** is to analyze healthcare data using **Power BI** to understand patient information, medical conditions, treatment patterns, hospital performance, and healthcare outcomes.

Key Objective:

Analyze Patient Demographics

Monitor Medical Conditions

Analyze Hospital Performance

Track Health Cost

Dataset Description:

Number of tables – 7

Tools Used:

PowerBI, PowerQuery, DAX, Microsoft Excel

Data Cleaning:

Removed duplicates

Handled missing values

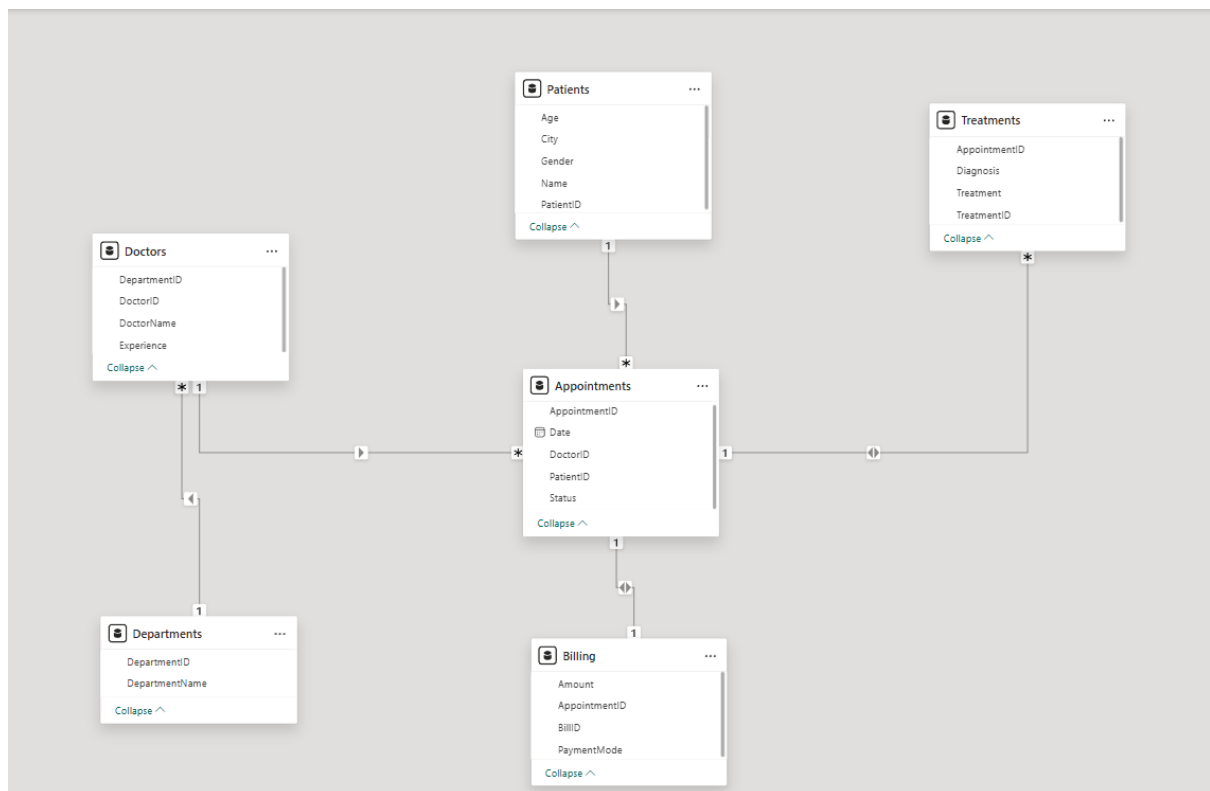
Changed data types

Cleaned column names

Transformed data

Created required columns

Data Modelling:



Patients[patient ID] (1)→ Appointments[Patient ID](*)Doctor[Doctor ID](1) → Appointments[Doctor ID](*)

Departments[Department ID](1) →
Doctors[Department ID](*)

DAX Measures:

Total Patients

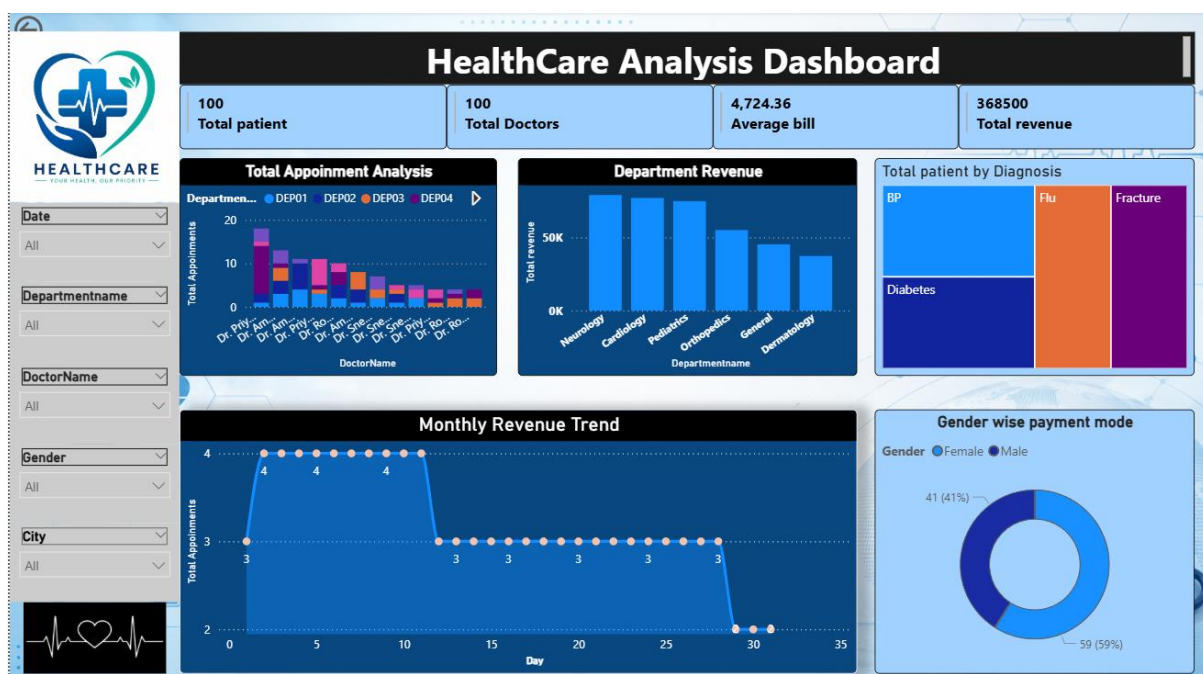
Total Doctors

Total Appointments

Total Revenue

Average Bill

Dashboard:



Key Insights:

1. Patient Insights

- Analyze the total patient population.
- Understand patient distribution by gender, city, and age.
- Identify which demographic groups contribute most to healthcare activity.
- Gender-wise analysis can help compare patient volumes.

2. Appointment Insights

- Analyze the total number of appointments.
- Track appointment activity by day/month.
- Analyze appointment status such as completed, cancelled, or pending, where available.
- Identify periods with higher or lower appointment activity.

3. Department Insights

- Compare revenue across departments.

- Analyze appointment activity by department.
- Identify departments with higher patient/treatment activity.
- Connect department performance with doctor activity.

4. **Doctor Performance**

- Compare doctors based on appointment volume.
- Analyze doctor experience.
- Evaluate revenue generated through doctor activity.
- Identify how appointments are distributed among doctors.

5. **Diagnosis & Treatment Insights**

- Identify the **most common diagnoses**.
- Analyze the distribution of treatments.
- Compare diagnoses across patients or departments.

- Understand which medical conditions contribute most to treatment activity.

6. 💰 **Financial Insights**

- Calculate **Total Revenue**.
- Calculate **Average Bill**.
- Analyze revenue trends over time.
- Compare revenue by department.
- Analyze payment methods such as the available PaymentMode categories.

Overall Insights:

Patient perspective: Understand patient demographics such as gender, age, and city.

Appointment perspective: Monitor appointment volume and trends over time.

Doctor perspective: Compare doctor activity and experience.

Department perspective: Analyze department-level activity and revenue.

Treatment perspective: Identify diagnosis and treatment patterns.

Financial perspective: Monitor total revenue, average bill, and payment methods.

Trend perspective: Track how appointments and revenue change over time.